DO NOW — RECALL

Linear search: trace and code

Name: _____ Class: _____ Date: _____

1. What property must both inputs satisfy before a standard merge step?

2. What does the key represent in insertion sort?

3. What is the last index of a 5-element C# array?

4. A loop checks a[0], a[1], a[2] then returns. How many comparisons were made?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Each input run must already be sorted.

2. Model answer

The value being inserted into the sorted prefix.

3. Model answer

4. Indexes are 0 through 4.

4. Model answer

3 comparisons.

Vocabulary — know these terms

target — Value the search seeks.

index — Zero-based position of an array element.

sentinel — Special result such as -1 meaning not found.

early return — Leave the function as soon as a match is found.

O(n) — Search may inspect every item as input grows.

NEW LEARNING

Read, annotate and explain

Begin at index zero

Linear search examines one value at a time, in order.

Stop at the first match

When a[i] equals target, return i immediately.

No match

After scanning every element, return -1; no ordering assumption is needed.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find 16 in [6,11,16,21]. Give the index and number of comparisons.

Predict first

Write visited indexes before identifying the match.

Model solution

Visit 0 (6), 1 (11), 2 (16). Return index 2 after three comparisons.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Search for 7 in [4,7,9,12]. Give index and comparison count.

Question 2

Search for 5 in [3,8,11,16]. Give return value and count.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Modify the function to count comparisons without returning early; explain the output for a 5-element array.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Linear search requires sorted input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.